

Processes soft modeling based on stacked autoencoders and wavelet extreme learning machine for aluminum plant-wide application

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ABSTRACT

Data-driven soft modeling has been extensively used for industrial processes to estimate key quality indicators which are hard to measure by some physical devices. However, the existing deep soft methods faces the challenge of training efficiency, gradient diminishing and explosion. Constructing an accurate and robust soft model is still a challenging topic from an application point of view. This paper develops an effective and efficient soft method (SAE-WELM) for processes modeling. First, a stacked autoencoder (SAE) is used to extract the deep features. Then, a top-layer extreme learning machine (ELM) is further applied to a plant-wide industrial aluminum production process. The activation function is wavelet kernel. Finally, the approximation and convergence of the proposed SAE-WELM are theoretically proved. The industrial case demonstrates that SAE-WELM captures the deep features faster than other iterative-based neural networks, and the accuracy and robustness outperform the existing state-of-the-art methods.

1. Introduction

In modern processes industry, some key quality variables are hard to measure because it suffers the environmental limitations, expensive offline cost, and large delay. Thus many data-driven soft strategies have been proposed, see for instance (Cong & Yu, 2018; Rengasamy, Jafari, Rothwell, et al., 2020; Xie, Hao, & Huang, 2019; Yan, Tang, & Lin, 2017; Yuan, Li, & Shardt, 2020; Zhu, Ge, & Song, 2017). The soft methods are mainly classified into three categories including black-box models, white-box models, and hybrid models (combining the black-and-white models). The white-box models are based on the first principles which limits the applications because it relies on a knowledge of the process mechanism. With the development of a distributed control system, data-driven soft 'black' models gradually gain more popularity because it is more flexible, for example, with some historical data one can train a predictive model (Le & Ge, 2017). Researchers are likely to explore the nonlinear and dynamic function projection between the input variables and output key quality as an economic alternative to the complex and sophisticated process knowledge (Shen, Yao, & Ge, 2020; Yuan, Ou, Wang, et al., 2019; Yuan, Qi, Wang, et al., 2020; Yuan, Zhou, Huang, et al., 2020).

Most of the existing soft models utilize different nonlinear architecture to extract information contained in the processes data. Some popular models include principle component analysis (PCA) (Zhu et al.,

2017), deep belief network (DBN) (Wang, Pan, & Yuan, 2020), artificial neural network (ANN) (Xie, Xie, & Ying, 2019), and probabilistic principal component regression (PPCR). Recently, some novel time-sequential models have been proposed. For example, Yuan et al. proposed a soft sensor model based on supervised long-short time memories (LSTM) (Yuan, Li, & Shardt, 2020; Yuan, Li, & Wang, 2019), the application results showed that the supervised training performance is superior to the traditional fine-tuning approach. Besides, Xie et al. developed a data-driven gated recurrent unit for soft sensor processing (Xie, Xie, & Ying, 2019). Two stream factors are introduced in a gated recurrent neural network, the simulation study illustrated a good performance result (Xie, Xie, & Ying, 2019). Ge et al. created an efficient soft model with some process regression tasks (Zhu et al., 2017). Chou discussed the development of a physically soft sensor using sequence-to-sequence models (Chou, Wu, & Kang, 2019). Jrvisalo built a soft sensor model inflow rate prediction for specific energy estimation (Jrvisalo, Ahonen, & Ahola, 2016). Then, (Yuan, Ou, et al., 2019) introduced the hierarchical representation method for soft sensor domains. They employed an improved gradient descent to update the learning parameters. Other time sequence models of industrial processes have been discussed in (Sun & Ge, 2018; Yuan, Qi, et al., 2020).

Currently, with the deep architecture development, the performance of soft models based on deep learning have been greatly improved.

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Those models utilize the deep representation ability of input variables instead of shallow architecture, the learning ability of complex tasks has been experimentally guaranteed. Further, with the sufficient processes, the hard environment and large amount of feature input in the processes industry, the deep learning techniques have shown its advantages. To obtain the complex nonlinear features, deep learning is considered as a remarkable frame to deal with these issues, and high accuracy and performance have been achieved (Duan, Dai, & Chang, 2015; Kols & Stre, 2009; Rafiei, Niknam, & Khooban, 2016; Sun & Ge, 2018; Xie, Xie, Li, et al., 2017; Yang, Dillon, Chang, & Chen, 2019; Yin, Niu, He, et al., 2020; Yuan, Li, Shardt, Wang, & Yang, 2020; Yuan, Zhou, & Huang, 2019; Zheng, Li, & Li, 2013). The deep learning approaches have been extended to industrial process optimization, monitoring, fault diagnosis, biology, and chemistry domains. Compared to the shallow networks, the complex feature representation ability is limited, and the generalization ability is not appropriate to large processes systems (Lei, Chen, Min, & Xie, 2020; Lei, Chen, & Xie, 2020; Yuan, Li, & Shardt, 2020). The aforementioned results give us the motivation and inspiration to use the deep technique into the aluminum industry. However, these deep neural networks are mainly based on the background propagation solution, gradient diminishing and exploding issues are challenging in the training process. The computational complexity of these iterative methods have difficulties with generalization.

To improve the computational efficiency in the single-hidden-layer neural network, Huang et al. proposed an extreme learning machine (ELM) with high performance in training the single-hidden layer network (Tang, Deng, & Huang, 2015a). ELM has been applied to many domains such as classification, regression tasks, pattern recognition, and industrial process domains. Many improved ELM-based methods have also been built. For example, Wong et al. designed a multi-layer kernel-based ELM for representing features (Wong, Vong, & Wong, 2016). Their model used different kernels and gained a generalized optimal model. Song et al. (Huang et al., 2017) proposed a semi-supervised ELM method which combines the unlabeled data for the training process (Huang et al., 2017). Since then, some semi-supervised ELM based methods are also extended to soft sensor domains (Le & Ge, 2017; Zhu et al., 2017). Lu developed an online spatiotemporal ELM for complex time-varying distributed parameter systems, which can accurately model time-varying and nonlinear industrial processes (Huang et al., 2017; Lu, Yin, Liu, et al., 2017). To take advantage of the deep feature extraction ability, the proposed method built the deep architectures for an aluminum electrolysis process. However, the traditional deep learning techniques use the back-propagation algorithms to train the whole network, it has the shortcoming of the gradient diminishing and exploration, so in our network, we use the stacked autoencoders for the complex feature extraction, and a top-layer extreme learning machine is used to the key quality identification. ELM-based training method is also used for training the proposed soft model layer-by-layer. Some existing models use the different activations to achieve the better performances.

Deep kernel learning has recently attracted increasing attention in the industrial process modeling applications (Huang, Oh, & Pedrycz, 2016). The main category of the deep kernel learning are: (1) Single kernel model; this model uses the single kernel as the nonlinear activation function, such as Sigmoid kernel, Gaussian kernel, and Tanh kernel. (2) Ensemble kernel model; this category integrated two or more kernels to the deep network architectures. Radial basis function (RBF) is one of the most widely-used models for learning the mapping function from the input to the output (Park and Sandberg). The common nonlinear activation function in RBF is the Gaussian kernel. However, the performance of RBF-based algorithms is largely decreased if the geometry distribution of input samples is not the Gaussian. Further, the long tail in the Gaussian activation function from minus infinity to plus infinity limits the denoising ability. Some frequency of the input variables is also discarded. In contrast to the Gaussian kernel

of RBF, wavelet neural network is a highly efficient method for signal processing, and a shallow neural network has been proposed with the wavelet activation function. The shift factor enables the time axis of the wavelet neural network to perform ergodic analysis along with the temporal features, and scale parameter can shrink and stretch the wavelets to approximate the signals at different frequencies. Therefore, with different scale and shift transform, the better time–frequency localization property and focal characteristics are achieved. For example, Huang et al. improved the traditional wavelet function and proposed a fuzzy wavelet neural network for prediction tasks (Huang et al., 2016; Huang, Oh, & Pedrycz, 2017). In (Rafiei, Niknam, & Aghaei, 2018), an improved wavelet neural network was built for electronic price prediction. This method considered the uncertainty factor and a bootstrapping technique are introduced, however, the accuracy is easily influenced by different noises. Zhang et al. mathematically analyzed the function can learn the nonlinear function which has a better convergence rate and universal approximation ability (Zhang, Walter, & Miao, 1995). (Ganjefar, Tofighi, & Karami, 2015; Rafiei et al., 2018; Xie, Xie, Gui, & Yang, 2018; Yilmaz & Oysal, 2010a, 2010b) give the recent development while their applications in the industrial plant process are not completely addressed, which partly motivates us for the current study.

To address the aforementioned problems, we proposed a hierarchical soft model to extract information and achieve a quality prediction for industrial processes. Wavelet activation was used to get the time-frequency location property of the input variables. First, the stacked autoencoder was used for the complex feature extraction. Second, an ELM with a wavelet kernel activation function is built for a topmost classifier. Through this innovative combination, a more reliable and robust soft model can be generated. The different activation function is experimented for evaluating the performance of this algorithm. The main contributions of the paper are given as follows:

1. The traditional deep architectures contain the issue of gradient diminishing and exploding. To improve the training efficiency and generalization ability, a novel deep soft model based on ELM with a wavelet kernel function is proposed.
2. A data-driven soft model was developed which can preserve the universal approximation ability and extract the deep and complex features in-process data. In contrast to the Gaussian kernel, the proposed approach combines the wavelet kernel in ELM, the generalization ability and training speed can be greatly improved.
3. To obtain a stable and applicable model, the approximation ability and convergence of the proposed SAE-WELM are proved by constructing a Lyapunov function.

The reminder of this paper is organized as follows. In Section 2, we give a description of ELM, stacked autoencoder, and some kernel activation functions. In Section 3, we introduce the LHSS-HELM formation. Public experiment evaluation and an industrial application of soft sensor technique are described in Section 4. Section 5 concludes the paper.

2. Preliminary foundations

2.1. Extreme learning machine

ELM is proposed in (Huang et al. 2015) which has been considered as an efficient and effective method to train the single-hidden layer feedback neural network. The core idea in ELM is that a Moore–Penrose solution is used as an alternative to the gradient descent process. It is proved that ELM has a fast computation and high generalization ability. Given a training sample set $\{x_i, t_i\}_{i=1}^N$, ELM constructs a network with two stages. At the first stage, we choose the randomly generated input

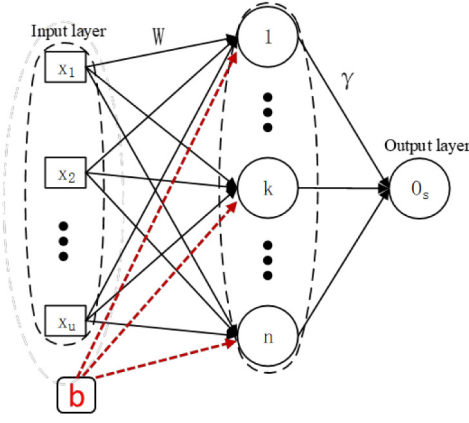


Fig. 1. Structure of the single-hidden layer network.

weights from a uniform distribution on $(-1,1)$. The bias b is initialized to a constant so the output of the hidden layer is represented as:

$$H = \begin{pmatrix} G(w_1, b_1, x_1) & \dots & G(w_1, b_L, x_N) \\ \vdots & \ddots & \vdots \\ G(w_L, b_1, x_N) & \dots & G(w_L, b_L, x_N) \end{pmatrix} \quad (1)$$

where L is the number of neurons in the hidden layer of the neural network. $G(\cdot)$ is the activation kernel function, which usually is a nonlinear and piecewise function (see Fig. 1). The single hidden-layer feedforward neural network can be described as:

$$\sum_{i=1}^N \beta g(w x_i + b) = y_i \quad (2)$$

The learning approximation function with N samples are:

$$\min_{\beta} \left\| \sum_{i=1}^N \beta g(w x_i + b) - y_i \right\| \quad (3)$$

So a commonly least square objective function can be built by the following equation.

$$H\beta = Y \quad (4)$$

Its output weight solution can be given in the following Moore-inverse formulation:

$$\beta = H^{\dagger} \cdot Y \quad (5)$$

Which is a closed solution of the hidden weight β rather than an iterative training, so the time complexity will be greatly improved.

2.2. Autoencoder

Autoencoder (AE) is an unsupervised single-hidden-layer neural network that maps the output layer equal to the input layer as accurately as possible. AE constructs a 3-layer neural network for fine-tuning of the optimal parameters. Differently than the radial basis network, the output of AE can be the same as the input. Assume that the input dimension of autoencoder is $x = [x_1, x_2, \dots, x_{(n_i)}]^T$, where n_i is the dimension of the input variables. First, the autoencoder projects from the input layer to the hidden layer by the mapping function as follows:

$$H = f(Wx + b) \quad (6)$$

where $H = [h_{(1)}, h_{(2)}, \dots, h_{(n_h)}]^T$ is the hidden representation and n_h is the dimension of hidden neurons. W is a weight index matrix with the

dimension of $n_i \times n_h$ and b is the bias vector. The cost function of an autoencoder can be expressed as:

$$\begin{aligned} L_{AE} &= \frac{1}{N} \sum_{i=1}^N |x - \hat{x}|^2 \\ &= \frac{1}{N} \sum_{i=1}^N |x - g(a \cdot x + b)|^2 \end{aligned} \quad (7)$$

where N is the input and $g(\cdot)$ is the related activation function. The loss function is solved by a gradient descent algorithm. Some variety of AE has been built such as denoising autoencoder and variational autoencoder. To obtain the deep feature representation and dimension reduction, the basic autoencoder can be overlapped layer-by-layer.

2.3. Kernel activation function

In the feedback neural network, there are several kinds of activation functions to be used for the hidden layer projection. The commonly used in FNN is the Sigmoid activation function and Tanh activation function. These two types of activation functions can be described, respectively, as:

$$g_1(a, b, x) = \tanh(x) = \frac{\exp(a \cdot x + b) - \exp^{-(a \cdot x + b)}}{\exp(a \cdot x + b) + \exp^{-(a \cdot x + b)}} \quad (8)$$

$$g_2(a, b, x) = \frac{1}{1 + \exp(-(a \cdot x + b))} \quad (9)$$

and also, the commonly used kernel in an RBF network is Gaussian activation kernel, which are described in the following equation:

$$g_3(a, b, x) = \exp\left(-\frac{b\|x - a\|^2}{2}\right) \quad (10)$$

where a is the weight parameter and b is the bias. The standard Sigmoid activation function cannot accurately build a projection mapping between the various samples and origin input signals (Lei, He, & Zi, 2011). The Gaussian kernel cannot deal with the noisy samples and only time characteristics are analyzed. The long-tail of Gaussian makes it hard to deal with the samples with high noises, and only with the Gaussian distribution that the model can gain a satisfactory performance.

3. Main methodologies

To address the issue of nonlinear mapping and achieve a better generalization ability, a novel soft sensor model based on wavelet activation function in ELM, so-called SAE-WELM, is developed.

3.1. Wavenet kernel transform

The wavelet transform is based on some little functions, called wavelets. The objective of the wavelet is to generate a signal approximation by some translated and dilated waves. The wavelet proved that it has a good frequency and temporal analysis, which means it can shrink very quickly in the high frequency thus a fast convergence rate can be obtained. A continuous wavelet is described as follows:

$$\Psi(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} f(t) \varphi\left(\frac{y-b}{a}\right) dt \quad (11)$$

where φ denotes the mother wavelet function. a and b are the variation of frequency and time parameters, respectively. As stated in (Zhang et al., 1995), the wavelet function can be used as a kernel function for training the single-hidden layer neural network. The representation of wavelet can be expressed as:

$$\hat{y} = \sum_{i=1}^L \beta_i \varphi_i \quad (12)$$

where $\hat{y}$ is the predicted value of the neural output. L is the number of hidden neurons in the hidden layer and β_i denotes the i th output

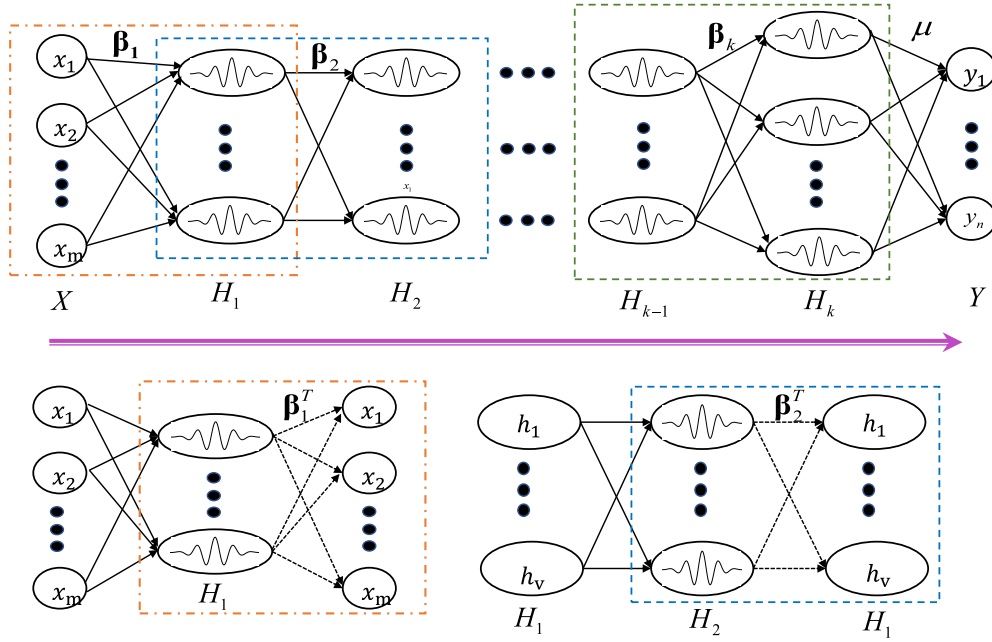


Fig. 2. The proposed framework of SAE-WELM.

weight connect the hidden layer and the output layer. The cost function of wavelet neural network is described in the following equation:

$$U_{loss} = \min_{\beta, a, b} \frac{1}{N} \sum_{i=1}^N (Y - \hat{Y})^2 \quad (13)$$

$$= \min_{\beta, a, b} \frac{1}{N} \sum_{i=1}^N \left(Y - \sum_{j=1}^L \beta_j \varphi_j(x_i) \right)^2$$

where $\hat{Y}$ is the predicted output and Y is the real labeled sample. β_j is the output weights of the j th hidden layer. Zhang proposed some heuristic algorithms for fine-tuning the weight vector. For example, the gradient descent algorithms and least mean squares (LMS) can be used to tune the weight parameters.

3.2. SAE-WELM model implementation

In this part, a novel deep SAE-WELM method is proposed. We use (X, Y) to denote the training samples. As shown in Fig. 2, a hierarchical autoencoder and ELM for the top-level is constructed. The present input of the SAE-WELM is equal to the input data of X . The i th hidden layer output is further for the $(i + 1)$ th output, and the whole network can be trained layer-by-layer. Differently from the traditional deep neural network, the continuous nonlinear wavelet activation function is applied. The deep SAE-WELM projects the input variables to a different dimension space using the following equation:

$$h_i = \beta \cdot g(\varphi(x)) + b_i, \quad i = 1, \dots, M \quad (14)$$

where M is the number of the hidden layers. The output matrix for the whole samples is presented by:

$$H = \begin{pmatrix} g(\varphi_{a_1, b_1}(x_1)) & \dots & g(\varphi_{a_L, b_L}(x_1)) \\ \vdots & \ddots & \vdots \\ g(\varphi_{a_1, b_1}(x_N)) & \dots & g(\varphi_{a_L, b_L}(x_N)) \end{pmatrix} \quad (15)$$

where a is the scaling translation and b is the dilation parameter. x is the number of training samples. N denotes the number of input nodes and L is the number of hidden neurons.

The projection output matrix is represented as follows:

$$H_j = g((\beta_j)^T H_{j-1}), \quad i = 1, \dots, K. \quad (16)$$

The output weights β are computed by a Moore–Penrose which meet the following constraints:

$$\begin{aligned} \beta &= H^{-1} X \\ \text{s.t. } \beta \beta^T &= I \end{aligned} \quad (17)$$

where I is an identity matrix whose square elements are ‘1’s, and other elements are ‘0’s. The implementation of the SAE-WELM can be given in the following steps.

- (1) Choose the related input variables according to the experienced worker and operators. All the collected data are preprocessed by the normalization and standardization.
- (2) Divide the dataset patterns with Training, Validation and Testing samples.
- (3) Determine the architecture of the proposed SAE-WELM, the number of the hidden layer, and the neurons in each layer. Initializing the shift a , dilation parameters b ; the residual error threshold ε .
- (4) Train the hierarchical autoencoder layer-by-layer with the Eq. (16), and gain the K hidden layers output of $H(1) - H(K)$.
- (5) Train the top ELM with Eq. (17) for getting the output weights of the SAE-WELM.
- (6) Evaluate the performance of the proposed soft models with the testing patterns, if residual error is lower than the threshold, we give the projection of the SAE-WELM. Otherwise, updating dilation parameter with the candidates, and then back to Step (4).
- (7) Apply the trained SAE-WELM for key quality identification to get an output of the layers weights and nonlinear projection function.

3.3. Approximation capabilities of the SAE-WELM

In this part, we aim to prove the universal approximation and convergence. Let $L^2(X)$ be a spatial function of f , the $L^2(X)$ is dense on a compact set X in a d -dimension affine space R^d . The space meet the limitation that $|f|^2$ is square integrable, $\int_X |f(x)|^2 dx < \infty$. Denote the compact subset of V and U , $V, U \in L^2(X)$. The inner product can be given as:

$$\langle V, U \rangle = \int_X U(X) V(X) dX \quad (18)$$

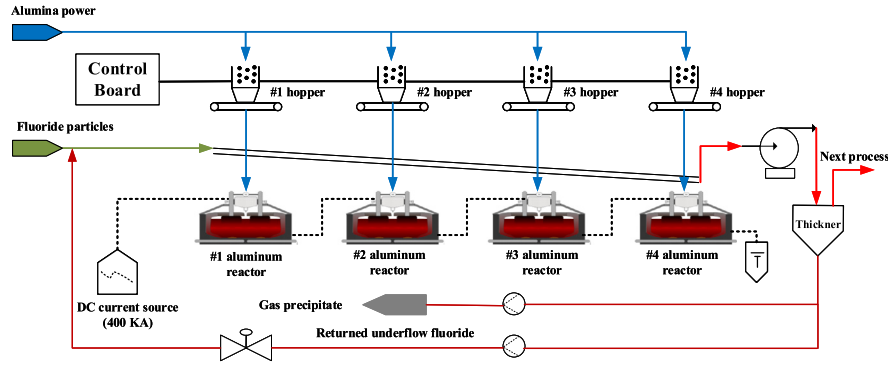


Fig. 3. Illustration of the aluminum reduction process.

Table 1

Training times and variances for different algorithms with different numbers of hidden nodes A: Variance, B: Training time (s), C: Testing RMSE.

N_h	ELM			RBF			SAE-WELM			DBN		
	A	B	C	A	B	C	A	B	C	A	B	C
100	0.0231	2.3	0.0891	0.1235	2.4	0.0812	0.0083	0.038	0.0076	0.0842	0.25	0.0282
200	0.0121	2.8	0.0742	0.0832	4.2	0.0563	0.0083	0.034	0.0023	0.0768	0.59	0.0187
400	0.0109	3.5	0.0653	0.0756	5.5	0.0481	0.0074	0.042	0.0215	0.0746	0.71	0.0153
600	0.0098	4.9	0.0649	0.0678	6.3	0.0257	0.0071	0.047	0.0157	0.0654	0.93	0.0125

So that the density between the prediction and actual output function is defined by:

$$\|\hat{f} - f\|^2 = \int_X |\hat{f} - f|^2 dX \quad (19)$$

A lemma is given in the following (Park & Sandberg, 1993):

Lemma 1. Given a bounded piecewise and continuous nonlinear affine function φ :

$$\int_{R^d} \varphi(x) dx \neq 0 \quad (20)$$

Then the span $\left\{ \varphi\left(\frac{x-a}{b}\right) : a, b \in R^d \times R^+ \right\}$ is dense in all space of $L^p(X)$ for every $p \in [1, +\infty]$.

Theorem 1. Given a nonlinear piecewise continuous wavelet function $\varphi(x)$, where $\varphi(x) \in L^2(x)$ is dense in spatial space $x \in R^d$. The residual error function in this architecture $g : R^d \rightarrow R$ satisfies:

$$\int_R g(x) dx \neq 0 \quad (21)$$

Then, for an integer $N > 0$, we can find:

$$\begin{aligned} \lim_{n \rightarrow \infty} \|e_n\| &= \lim_{n \rightarrow \infty} \|g - f\| \\ &= \lim_{n \rightarrow \infty} \left\| \sum_{i=1}^L \beta_i \varphi_i(x) \right\| = 0 \end{aligned} \quad (22)$$

Proof. According to Lemma 1, $e_n \in L^2(x)$. Let $V(n) = \|e_n\|^2$ be a Lyapunov function, then the residual of error is defined:

$$\begin{aligned} \Delta &= V(n-1) - V(n) \\ &= \|e_{n-1}\|^2 - \|e_n\|^2 \\ &= \|e_{n-1}\|^2 - \|e_{n-1} - \beta_n \varphi_n(a, b)\|^2 \\ &= 2\beta_n \langle e_{n-1}, \varphi_n(a, b) \rangle - \beta_n^2 \|\varphi_n(a, b)\|^2 \end{aligned} \quad (23)$$

The above equation satisfies that:

$$\begin{aligned} \nabla &= \|e_{n-1} - e_n\|^2 \\ &= \|e_{n-1}\|^2 - 2\langle e_{n-1}, e_n \rangle + \|e_n\|^2 \\ &= \|e_{n-1}\|^2 - 2(\langle e_{n-1}, e_n \rangle - \langle e_{n-1} - e_n, e_n \rangle) + \|e_n\|^2 \end{aligned} \quad (24)$$

Table 2

The structure parameters of the different algorithms.

Methods	(K)	i th (V_i)	Hidden units	Initialization
ELM	—	—	200	$C = 25$
ANN	—	—	200	Learning rate 0.02
RBF	—	—	200	Learning rate 0.02
DBN	3	10	200	—
This paper	3	10	200	$a = 4.5, b = 0$

as stated in Tang, Deng, and Huang (2015b), then (24) holds that:

$$\begin{aligned} \nabla &= \|e_{n-1} - e_n\|^2 \\ &= \|e_{n-1}\|^2 - \|e_n\|^2 = (e_{n-1} + e_n)(e_{n-1} - e_n) \\ &= V(n-1) - V(n) \end{aligned} \quad (25)$$

Due to the condition that $\lim_{n \rightarrow \infty} \|\beta_n \varphi_n(a, b)\|^2 = 0$, then:

$$\lim_{n \rightarrow \infty} \|e_{n-1} - e_n\|^2 = \lim_{n \rightarrow \infty} (\|e_{n-1}\|^2 - \|e_n\|^2) \quad (26)$$

With the constraint of (23), (26) can be replaced by the following equation:

$$\begin{aligned} \lim_{n \rightarrow \infty} \|e_{n-1} - e_n\|^2 &= \lim_{n \rightarrow \infty} \|V(n-1) - V(n)\| \\ &= \lim_{n \rightarrow \infty} (\|e_{n-1}\|^2 - \|e_n\|^2) \\ &= \lim_{n \rightarrow \infty} (\|e_{n-1}\|^2 - \|e_{n-1} - \beta_n \varphi_n(a, b)\|^2) \\ &= \lim_{n \rightarrow \infty} (2\beta_n \langle e_{n-1}, \varphi_n(a, b) \rangle - \beta_n^2 \|\varphi_n(a, b)\|^2) \\ &= \lim_{n \rightarrow \infty} \|\beta_n \varphi_n(a, b)\|^2 \\ &= 0 \end{aligned} \quad (27)$$

Therefore, the convergence of the wavelet extreme learning machine can be guaranteed.

Remark 1. This proof shows that in a compact and dense subset space, the wavelet kernel can approximate any nonlinear continuous function. The residual error of the whole network will tend to zero when the number of hidden neurons of the proposed architecture equals to infinity. The theoretical proof of a robust model for a plant-wide application is given.

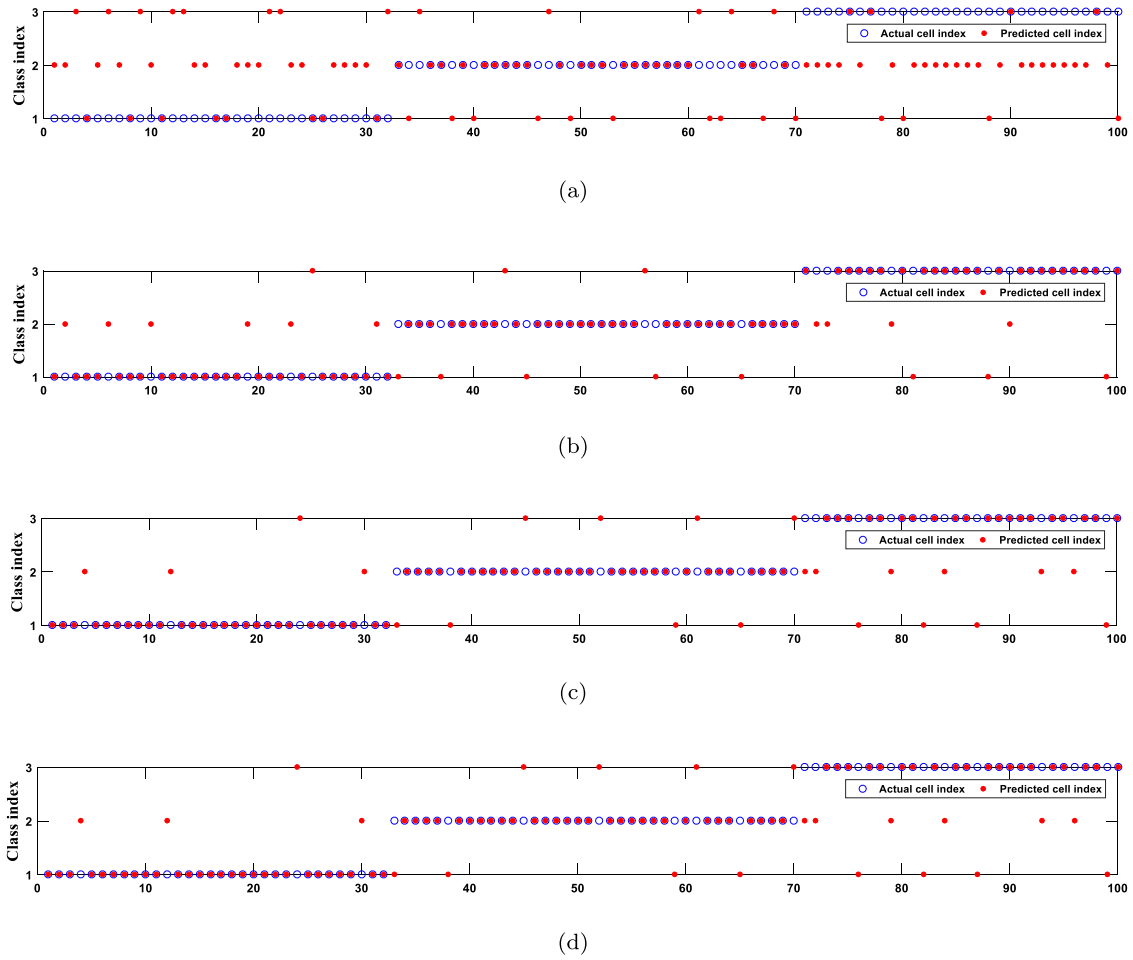


Fig. 4. Performance comparisons of different methods: (a) ANN; (b) SVM; (c) ELM; (d) HELM. If the blue circle overlaps the red point, which represents the right label prediction of SD. The corresponding accuracy is 35%, 42%, 67%, 88%, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Case study: Aluminum electrolysis plant experiments

4.1. Process description

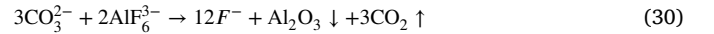
In industrial aluminum production, superheat degree (SD) is the difference of cell temperature and crystal situation. It is a very important indicator for monitoring the whole cell situation, improving the quality of aluminum and leveraging the electronic power cost. However, some key parameters cannot be directly measured because of the influence of hard environment, high temperature, and strong magnet. In practical aluminum production, the crystal temperature requires an off-line laboratory analysis to get an accurate value, which improved the detection cost of SD. From the knowledge of the aluminum process reaction mechanism, some inner link exists between these variables and the output key variable, which give us the insight to apply the proposed SAE-WELM for soft modeling in industrial aluminum plant.

Fig. 3 is the process flow diagram of the aluminum electrolysis process. The main reactions in the bath can be given as:

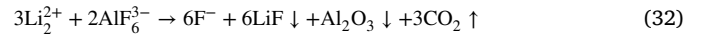


From the above two equations, the alumina (Al_2O_3) and the carbon anode are the main reaction raw materials in the aluminum reduction cell. However, the alumina power has a high liquidus temperature (up to 1011 °C), a high electrolytic energy will be wasted. To lower

the energy consuming, aluminum fluoride (AlF_3) and calcium fluoride (CaF_2) are added to the bath, hence some subsidiary reactions are:



and a total neutralization reaction is:



All of these complex reactions environments make it hard to direct the detection of the SD situation. Very often the SD detection needs two steps, the first stage requires a manual operator by using thermal device to get the bath temperature; the second stage is an off-line laboratory analysis of the crystal temperature. Both of the steps are economic and time consuming. In this part, we conducted the proposed SAE-WELM algorithm on an aluminum electrolysis processes for SD identification. SD situation has three situations, High, Middle, and Low. High label denotes the difference of cell temperature and crystal temperature is bigger than 15 °C. Middle label in the SD range is 8–12 °C, and the low label denotes the difference is lower than 8 °C.

4.2. Soft modeling setting and implementation

To extract complex industrial features in an efficient manner, an approximate input features can guarantee SAE-WELM's performance. According to the artificial experience, 13 selected input variables that is most related to the output quality are given in Table 3. In Le's

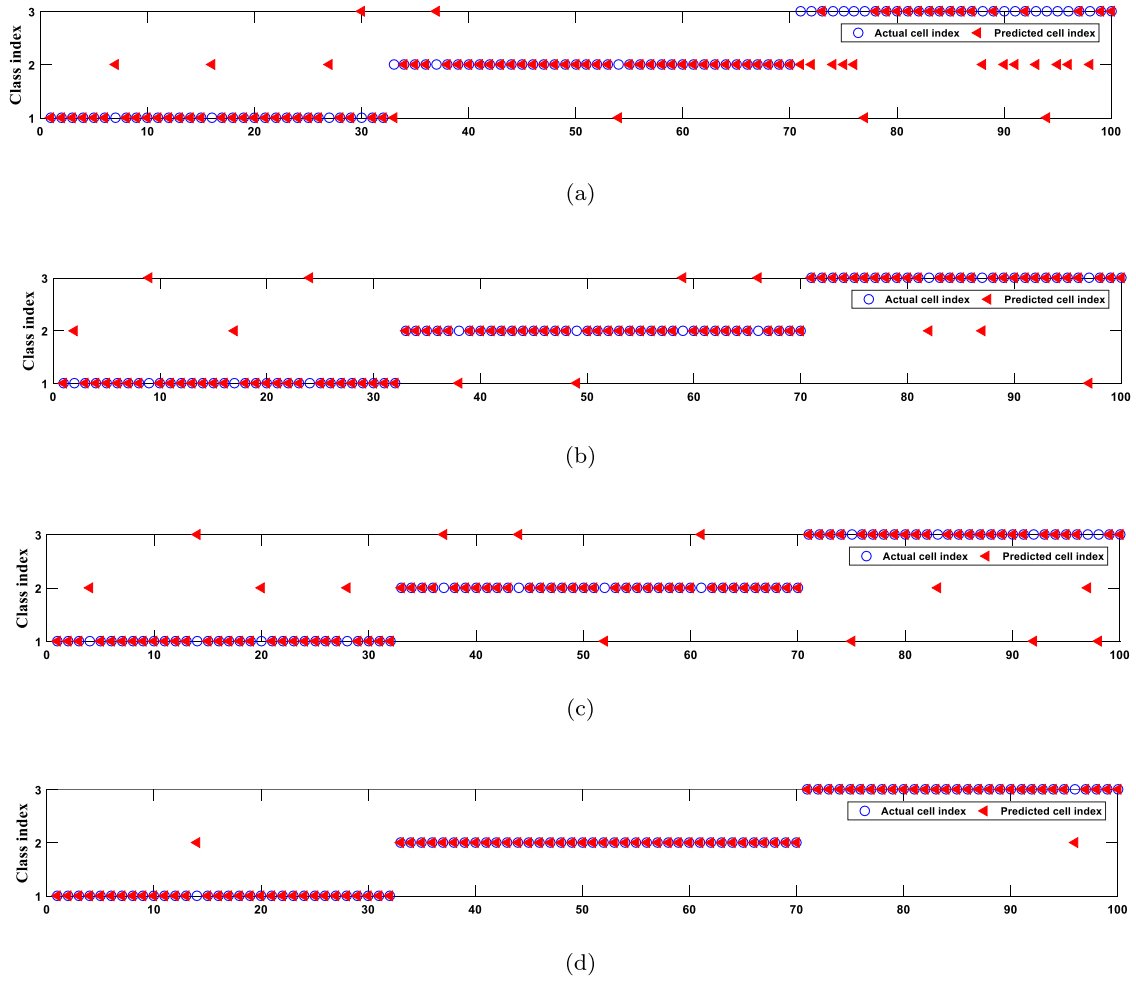


Fig. 5. Performance comparisons of different kernel of algorithms: (a) Gaussian RBF; (b) DBN; (c) Sigmoid (SAE-ELM); (d) Ours (SAE-WELM). The corresponding accuracy is 79%, 89%, 87%, 98%, respectively.

report ((Le & Ge, 2017)), the hidden neurons were set to 10 with a state-of-the-art performance, so in our experiment, the number of the hidden features are set to 10. Before conducting the SAE-WELM, 1000 patterns are collected. 100 samples are testing dataset and the 100 are validation, and the remaining are the training dataset. To make a competitive comparison, some popular kernel activation functions including the Sigmoid and Gaussian kernel are both conducted. Some persuasive methods such as SVM, RBF, Sig-ELM, DBN, ANN, and HELM ((Le & Ge, 2017)) are also compared to the experiments.

The proposed soft modeling ran in an PC with a 2.7 GHz CPU and 16 GB RAM in a Microsoft Windows 10 environment. The algorithm is implemented using Matlab 2016a. The hidden neurons in RBF and ANN network are set to 200 and the learning rate were set to 0.1. The root mean square error $RMSE$, training time (T) were used to evaluate the performance, computation cost, and robustness of the SAE-WELM soft processes modeling, respectively. $RMSE$ is defined as:

$$RMSE = \sqrt{\sum_{i=1}^N (y_i - \hat{y}_i)^2 / (N - 1)} \quad (33)$$

where y_i and $\hat{y}_i$ are the labeled and predicted output values of testing examples, respectively, and N is the number of testing samples.

Before training the competitive methods, Table 2 presents the parameters for the methods of ANN, ELM, RBF, DBN and our proposed methods.

Table 3

Selected input processes variables according to operator experiences.

No.	Tags	Descriptions
1	u_1	Working current of the whole bath
2	u_2	Cell resistance of the i th aluminum reactor
3	u_3	Set datum time of Al_2O_3
4	u_4	Actual feeding time of Al_2O_3 power of the bath
5	u_5	The content of Iron
6	u_6	The content of Silica
7	u_7	The aluminum tapping level
8	u_8	The electrolyte level
9	u_9	The precipitated aluminum amount
10	u_{10}	Electrolysis bath temperature
11	u_{11}	The ratio of molecular of the selected reactor
12	u_{12}	Average bath voltage
13	u_{13}	Additional amount of AlF_3 of the i th reactor

4.3. Experimental results

Some intuitive results are described in (Lei, Chen, et al., 2020; Lei, Chen, & Xie, 2020). From the 100 testing patterns performance comparison, the SAE-WELM have the significant accuracy improvement than the traditional ANN. The results demonstrates that some deep architecture (HELM, DBN) has a better representation learning ability than the shallow network (ANN).

Compared with Figs. 6 and 9, the average testing accuracy and training time of different algorithms with the different hidden units are given. The results show that the accuracy is relatively stable with the

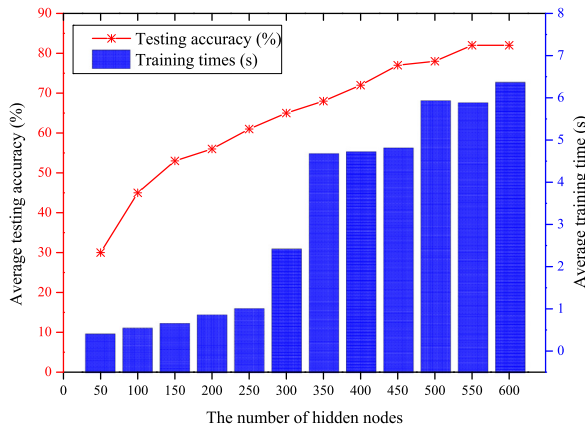


Fig. 6. Performance of ANN with different hidden-layer nodes.

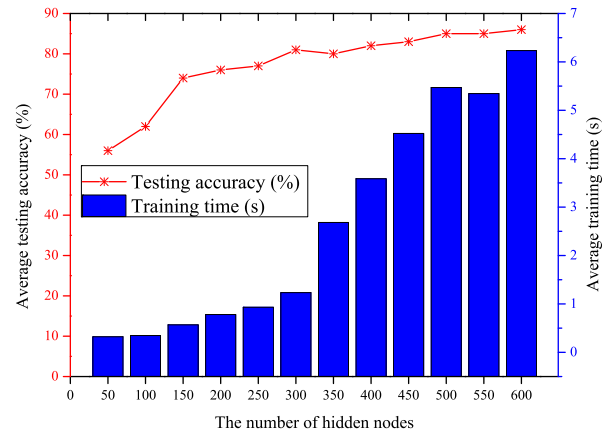


Fig. 7. Performance of RBF with different hidden-layer nodes.

advantage of a less training time, while RBF with Gaussian kernel is not robust than SAE-WELM. A low RMSE value indicates a superior performance (see Figs. 7 and 8). One can see that the accuracy of these methods are generally improved with the increment of the hidden units in both ANN and RBF. However, the training time is also an increasing tendency, which means more computation cost is necessary for building a high-dimension neural network. Due to the fast metric closed solution of the proposed SAE-WELM, the training speed is clearly better than BP-based ANN and RBF. A good performance can be shown in these figures. The ANN is the most time consuming method, which is the result of BP-based iterative steps. Our method is competitive in time complexity with RBF network, however, the accuracy outperforms the RBF based networks.

To make a more detailed robustness analysis of the proposed method, Table 1 presents the result with a different hidden layer units. To make a comparison, RBF, ANN and SAE-WELM are as a competitive rival. The intuitive shows that the proposed method is not sensitive to manual selection of user-specified parameters. From these experimental results, the proposed soft model has a good performance robustness in SD identification of the aluminum electrolysis plant. Table 1 gives the related index of variance to evaluate the robustness, compared to traditional ELM and RBF methods, the proposed SAE-WELM has the competitive advantage in robustness. Further, with the increase of the hidden nodes, the variance decreased, which means the proposed SAE-WELM has a better generalization performance.

4.4. Discussions

Based on the former experiments in the aluminum electrolysis plant application, the SAE-WELM has achieved the following significant performances.

Excellence performance in the wide-plant SD identification: Since the SD classification in industrial aluminum production is a challenging work for operators and technicians. An efficient and intelligent deep scheme is necessary to replace the manual detection which has the difficulty of subjectivity. The industrial experiments showed the proposed SAE-WELM provided a superior SD accuracy.

Reduction of the computation cost: The traditional methods are back-propagation based neural methods, which will have the issue of gradient diminishing and explosion. The proposed SAE-WELM is to learn the hidden weights by a closed matrix solution, so the computational efficiency is hundreds of times faster than the iterative approaches. The proposed method can be directly used for the plant SD classification as an alternative to the human, hence save cost and increase the profit.

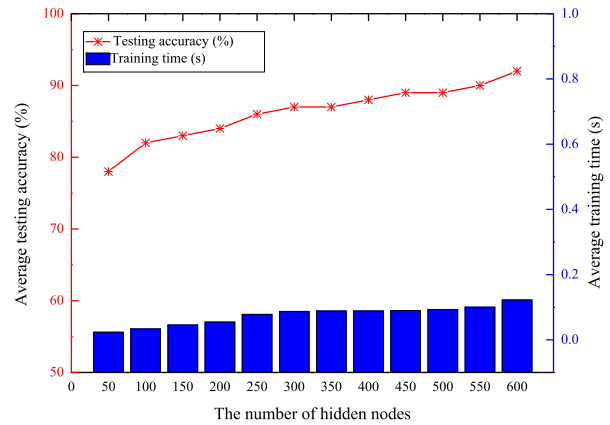


Fig. 8. Performance of the proposed method (SAE-WELM) with different hidden-layer nodes.

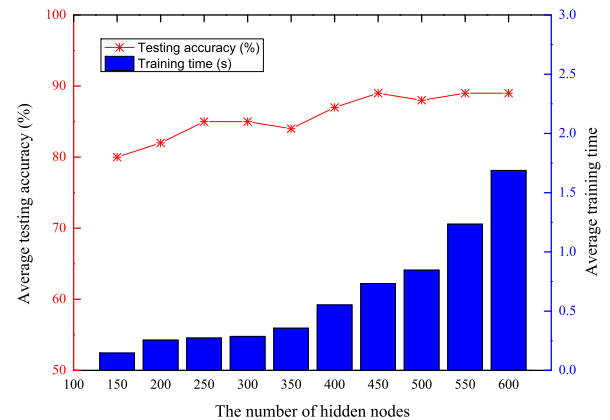


Fig. 9. Performance of DBN with different hidden-layer nodes.

5. Conclusion

In this paper, an effective and efficient soft model based on deep extreme learning machine and wavelet activation function was developed. This model reserves universal approximation ability and fast training speed. Industrial case in the aluminum plant-wide production demonstrated that the proposed SAE-WELM has good robustness and generalization ability. In principle, the proposed soft model can be extended to other plant-wide industrial process applications. Furthermore, the following topics deserve a further discussion,

(1) Wavelet is a stable bandwidth filter, some multi-resolution analysis (MRA) of the samples are deserving to be discussed. The time-frequency information of the related input variables can be used for training the SAE-WELM;

(2) The flexibility of the proposed method can be further discussed, an adaptive neural network can be built to optimize the proposed SAE-WELM architecture;

(3) The SAE-ELM performance relies on the manual feature selection. To deal with the noisy data, the performance of the proposed wavelet-based soft model can be further explored.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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